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Environmental regulation, emissions and productivity: Evidence from Chinese COD-emitting manufacturers☆

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ABSTRACT

In recent years, China's environmental regulation efforts have mainly focused on severely polluted "key regions." The central government has designated the "three rivers and three lakes basins" (3Rs3Ls) as key regions for water pollution control and has imposed a variety of regulations to improve water quality in those basins. This paper evaluates the effects of the water quality regulations on firms' emissions of chemical oxygen demand (COD) and productivity in the 3Rs3Ls basins. We find that although the water quality regulations forced many small, heavily-polluting firms to shut down, they had no statistically significant effects on surviving firms' productivity because they were ineffective in reducing their COD emissions. A policy that forces the surviving firms to reduce their emissions would reduce their output values and productivity, at least in the short run. However, the effect is likely to be small. Specifically, a 10% reduction in total COD emissions from the industrial sectors would require only a 0.1% reduction in output values under the current production technologies. These findings are robust to alternative specifications and sampling strategies.

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1. Introduction

A key question in environmental economics and policy is whether environmental regulation adversely affects firms' productivity. To address this question, the empirical literature usually regresses productivity on a measure of environmental regulation. A recent example is Greenstone et al. (2012), who estimated the effects of air quality regulations on manufacturing plants' total factor productivity (TFP) in the United States. A major challenge for this approach is to measure the stringency of environmental regulation properly (Brunel and Levinson, 2016). This is especially the case for many developing countries, where informal regulations exist and the compliance rate varies widely across firms for many reasons (Pargal and Wheeler, 1996). In another line of literature, studies estimate production functions in which emissions are modeled as an input for

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producing the desired output. Parametric and nonparametric models are proposed for this purpose (Cropper and Oates, 1992; Murty et al., 2012). One advantage of this approach is that it can be used to investigate the opportunity cost of emission reduction (in terms of loss of the desired output) under the current production technology. However, this approach has not been widely used in part because emission data at the firm level are not readily available.

The present paper contributes to the literature by empirically investigating the links between environmental regulation and firms' emissions, output, and productivity, using detailed firm-level data from China. Since 1996, China has gradually designated counties in the three rivers (i.e., Huai, Hai, and Liao) and three lakes (i.e., Tai, Chao, and Dianchi) basins (hereafter, 3Rs3Ls basins) as key regions for water pollution control and has imposed a variety of regulations to improve water quality in these basins. The regulations target specific manufacturing industries in each basin for water pollution reduction, and the targeted industries are subject to stricter environmental regulation. Chemical oxygen demand (COD) is a major type of water pollutants, which the Chinese government aims to control in its environmental regulation endeavors.[1]

An overarching finding of this paper is that the water quality regulations had no statistically significant effect on productivity of surviving firms in major COD-emitting industries in the 3Rs3Ls basins during the study period (1998–2007). There are two possible explanations for this result. One is that the regulations were ineffective in forcing firms to reduce their emissions and therefore did not affect their productivity. The second explanation is that environmental regulation does not necessarily reduce firms' productivity, as suggested by the well-known Porter hypothesis (Porter, 1991; Porter and van der Linde, 1995). Porter (1991) posits that emissions are a sign of waste of productive input; strict environmental regulation that promotes firms to use innovative approaches to reduce emissions may improve their productivity over time. To see which explanation is more plausible, we first estimate an emission function that links the level of COD emissions to a measure of water quality regulations in the 3Rs3Ls basins. This equation provides a direct test of the first explanation – if the water quality regulations are ineffective, they should not have much effect on individual firms' COD emissions. We then estimate a production function that takes emission as an input for producing the desired output. This equation provides evidence on the second explanation – if COD emissions are an inevitable by-product of producing the desired output under the current production technology, then efforts to reduce COD emissions will necessarily reduce the level of the desired output, at least in the short term.

Our empirical results provide strong evidence to support the first explanation. Although the water quality regulations in the 3Rs3Ls basins forced many small, heavily-polluting firms to shut down, they were ineffective in reducing surviving firms' COD emissions and, therefore, had no statistically significant effects on their productivity. Since 2007, the central government has increasingly shifted its regulatory and enforcement efforts to the larger surviving firms.[2] Our results suggest that this will decrease their output values, but not significantly. Specifically, a 10% reduction in total COD emissions, a goal specified in the central government's 13th Five-Year Plan (2016–2020), will require only a 0.1% reduction in output values under the current production technologies. These findings are robust to alternative specifications and sampling strategies.

The present paper makes three contributions to the literature. First, we believe this is the first study that uses data from a large sample of manufacturers to study water quality regulations and their impacts on firms' productivity in China. Most existing studies focus on environmental regulation in developed countries and, in many cases, reach opposite conclusions (see Ambec et al. (2013) or Jaffe et al. (1995) for reviews of related literature). For instance, Boyd and McClelland (1999) found that environmental constraints reduced the production level of a sample of US paper mills by 9%, of which 25% was the result of requirements for installing certain pollution abatement equipment. In contrast, Berman and Bui (2001) found a positive correlation between the air quality regulation and productivity of oil refineries in the Los Angeles Air Basin. Managi et al. (2005) found no statistically significant relationship between the stringency of environmental regulation and productivity of the oil and gas production sector in the Gulf of Mexico from the late 1960s to the late 1990s. Rassier and Earnhart (2010) studied the influence of the Clean Water Act on corporate profits in the United States and found a statistically significant negative effect. The present paper not only estimates the productivity impact of the water quality regulations in China, but also explores possible explanations for the results.

Second, our regulation measure varies by county, industry, and year. It is constructed based on the timing of designation of individual counties and lists of targeted industries for limiting COD emissions in each basin. To find out whether a particular 4-digit industry in a basin is subject to the water quality regulations, we examine various policy, technical, and statistical documents published by the Chinese government.

Third, this paper investigates the link between firms' output and emissions. We estimate a production function that treats emissions as an “input” (Copeland and Taylor, 2003; Cropper and Oates, 1992). This allows us to estimate the opportunity cost of COD emission reductions under the current production technologies. The results have clear policy implications as China's central government has initiated the “total quantity control of pollutants” policy that forces firms to reduce their emissions.

The remainder of the paper is organized as follows. Section 2 discusses the background of COD prevention and control in the 3Rs3Ls basins. Section 3 describes data sources for this study. Section 4 presents evidence of TFP impacts of the water

[1] See the next section of the present paper for a brief review of related policy on preventing and controlling COD emission in China.

[2] In 2007, the central government launched the National Program on Specially Monitored Firms (NPSMF), which explicitly targeted large firms. The program placed key industrial polluters under the central government's special monitoring. Among the 6066 key industrial polluters, there were 3115 water-polluting firms. According to the official document, the NPSMF firms were selected on the basis of their pollutant emissions over the last two years. The top ranked firms, which accounted for 65% of total emissions of either COD or NH3-N, were designated as the NPSMF water-polluting firms.

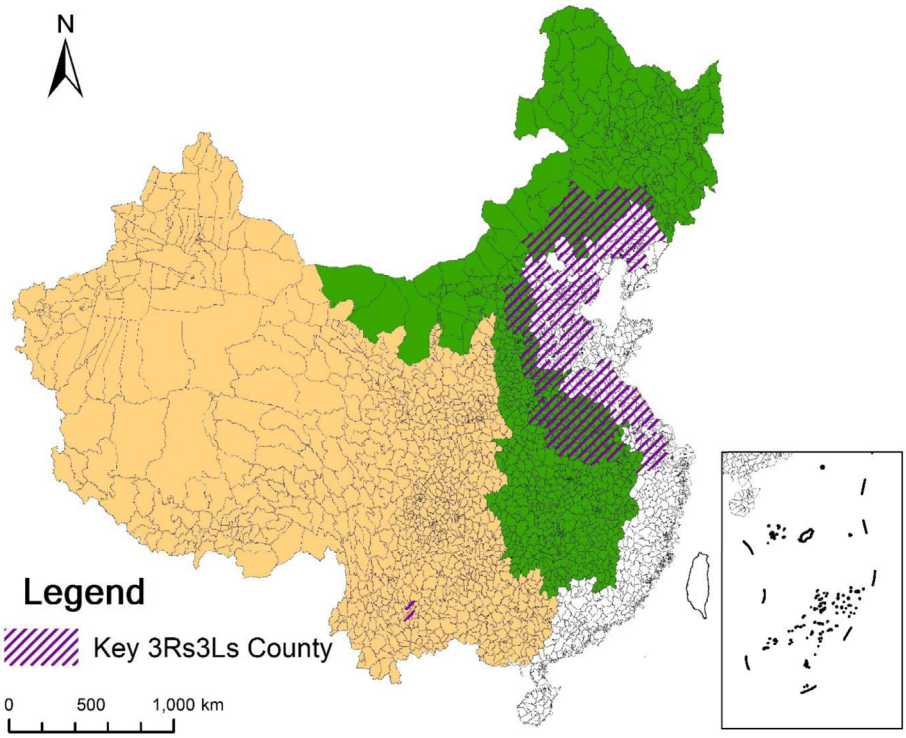

Fig. 1. Three rivers and three lakes regions, 2003.

quality regulations. Sections 5 and 6 explore possible explanations for the estimated TFP results. Section 7 analyzes heterogeneities and checks the robustness of the results. The final section concludes.

2. COD emissions and regulations in the 3Rs3Ls basins of China

Many rivers, lakes, and coastal waters in China are severely polluted as a result of agricultural, industrial, and domestic discharges. Major rivers are polluted by organic matter, while major lakes are also severely polluted by nitrogen and phosphorus, which leads to growing problems of eutrophication (OECD, 2007). Since 1996, the central government has identified the 3Rs3Ls basins as key regions for water pollution control.

2.1. Defining key regions

The State Environmental Protection Administration[3] was the principal government agency that developed the prevention-and-control plans for the 3Rs3Ls basins. The basins of the three rivers are located in northern China, while the basins of the three lakes extend over the southern part of the country (see Fig. 1).[4] The total drainage area of the 3Rs3Ls basins covers 810,000 km^2, traverses 14 provinces, and is inhabited by around 360 million people (OECD, 2007).

These prevention-and-control plans define the government's goals on water quality and total emissions of the regulated pollutants (e.g., COD), for each key basin, during the planned period, and set reduction targets for major pollution sources (e.g., agriculture, industry, and domestic sources). These plans also define the key regions, identify industries for regulation, and develop regulatory measures for achieving water quality goals.

As an important part of the prevention-and-control plans for the 3Rs3Ls basins, the government identifies key counties for limiting water pollution in each of the 3Rs3Ls basins. Table 1 documents the State Council's approval dates for the prevention-and-control plans for individual basins. In 1996, the Huai River basin was designated as a key region for water pollution control. The other five basins were designated in 1998 or 1999. A few more counties were later added to the list of key

[3] In March 2008, this agency was replaced by the Ministry of Environmental Protection.

[4] Fig. 1 shows the regulated 3Rs3Ls counties in 2003. This figure uses different background colors to indicate western, middle, and eastern parts of China. The criteria for classifying provinces into three different regions are found in public documents from the government. The eastern region includes the following provinces: Liaoning, Hebei, Tianjin, Beijing, Shandong, Jiangsu, Shanghai, Zhejiang, Fujian, Guangdong, Hainan, and Guangxi. Central: Heilongjiang, Jilin, Inner Mongolia, Shanxi, Anhui, Jiangxi, Hunan, Hubei, and Henan. Western: Chongqing, Sichuan, Yunnan, Guizhou, Shaanxi, Qinghai, Gansu, Ningxia, Xinjiang, and Tibet.

Table 1
Prevention-and-control plans for key drainage basins: Approval dates and industries.

Key region	Approval Date	Regulated Manufacturing Industries (CIC code)
Panel A: Plan for the 9th Five-Year period (1996–2000)		
Huai River	June 29, 1996	152, 22
Hai River	March 11, 1999	1462, 1469, 152, 1910, 192, 22, 262
Liao River	March 11, 1999	none
Tai Lake	January 6, 1998	22
Chao Lake	June 7, 1998	none
Dianchi Lake	September 6, 1998	none
Panel B: Plan for the 10th Five-Year period (2001–2005)		
Huai River	January 11, 2003	1391, 1462, 1469, 152, 171, 172, 174, 1910, 192, 22, 26, 27
Hai River	March 4, 2003	1462, 1469, 152, 171, 172, 174, 1910, 192, 22, 26, 27
Liao River	January 30, 2003	13, 14, 152, 171, 172, 174, 251, 22, 26, 27, 31, 32, 33
Tai Lake	August 31, 2001	171, 172, 174, 1910, 192, 26, 27
Chao Lake	December 23, 2002	13, 14, 17, 262, 263, 267, 30, 31
Dianchi Lake	March 12, 2003	26

Notes: 1. Data for this table are obtained from each basin's prevention-and-control plans for the given periods, and the State Council's approval of these plans. 2. The 3Rs3Ls plans for the 11th FY (2006–2010) were approved by the central government in April 2008, which is beyond the periods covered by the firm-level data. 3. The initial 1996–2000 plans for the Liao River, Chao Lake, and Dianchi Lake basins emphasized reduction in emissions from agricultural and domestic sources and did not list any manufacturing industries for COD emissions control. 4. Numbers in the last column are the Chinese Industry Classification (CIC) codes of the regulated industries.

counties in the basins of Huai River and Hai River. The designation of key counties was mostly based on the geographical location of the counties.

2.2. Identifying targeted industries

Reducing COD discharges has been a major focus of the water quality regulations in the 3Rs3Ls basins.[5] COD measures the oxygen consumed by chemical breakdown of organic and inorganic matter in water. It is used to indirectly determine the amount of organic compounds in water, which may have negative impacts on aquaculture production, human health, and ecosystem services. Industrial sectors are important sources of COD. In 2004, the sectors discharged 5.1 million tonnes of COD to receiving water bodies in China.[6] Major emitting industries include: processing of food from agricultural products (CIC 13), food (14), beverages (15), textile (17), paper and paper products (22), raw chemical materials and chemical products (26), medicines (27), and smelting and pressing of ferrous metals (32). In total, these eight industries account for more than 75% of total COD emission from industrial sectors in 2004 (see Table A1 in the appendix for details).

As an important part of the COD emissions control strategy, the prevention-and-control plans for the 3Rs3Ls basins list targeted COD-emitting industries for each basin. They are mainly at the 2-digit level, but there are also some at the 3-digit or 4-digit levels. The third column of Table 1 shows the targeted industries in each basin. Note that the shares of emissions from specific emitting industries vary widely across the 3Rs3Ls basins, due to geographical location of emitting firms from different industries. For example, textiles (CIC 17) accounted for about 48% of COD emissions in the Huai River basin in 2002, while smelting and pressing of ferrous metals (32) was the largest source of COD in the Chao Lake basin, accounting for nearly 40% of the total industrial discharge in this basin (see Table A2).

A comparison between the list of targeted industries in the prevention-and-control plans (as shown in Table 1) and the list of major polluting industries in each basin (as shown in Table A2 in the appendix) reveals that the plans for some basins list all COD emitting industries, including some that are barely present in the basins. To determine which industries were actually targeted for limiting COD emissions, we use the following procedures. First, we identify the 2-digit industries[7] that contain the listed industries in the prevention-and-control plans and account for 3.5%[8] or more of the total emissions in each basin. Second, heavy COD emitters are defined at the 4-digit level within the identified 2-digit industries. For that purpose, we use information from the *Handbook on Emission Coefficients of Industrial Sources of Pollution for the First National Census on Pollution Sources* (hereafter, the *Handbook*), which clearly identifies whether a 4-digit industry emits COD. Third, combining the information from the first two steps allows us to identify the regulated COD emitters at the 4-digit level for each of the 3Rs3Ls basins during each time period. Essentially, a 4-digit industry in a specific basin is treated as a regulated industry for COD reduction if it is 1) a heavy emitter as suggested by the *Handbook*, 2) a major contributor to the COD emissions in the

[5] Regulatory efforts in the basins also aim to reduce phosphorus (TP) and nitrogen (TN) runoffs into the lakes. The present paper focuses on the COD emissions as the manufacturing sector is not a major source for TP or TN.

[6] Domestic sources discharged 8.3 million tonnes of COD in 2004. In 2007, COD emissions from industrial and domestic sources are 5.1 and 8.7 million tonnes, respectively.

[7] Emission data are not available at more disaggregate levels.

[8] We use a threshold to define heavy emitters. In this paper, we mainly report the results that correspond to a threshold of 3.5%. Also, we use 10% threshold to do a robustness check and find similar results which are reported in Section 7.

Table 2
Regulated industries in the present study: 1998–2007.

Regions	Period	Regulated industries for limiting COD emissions
Huai River	1998–2002	152, 22
	2003–2007	1462, 1469, 152, 171, 172, 174, 22, 26
Liao River	1998–2002	none
	2003–2007	13, 152, 22, 26, 27, 32
Hai River	1998	none
	1999–2002	1462, 1469, 22, 262
	2003–2007	1462, 1469, 22, 26, 27
Tai Lake	1998–2001	22
	2002–2007	171, 172, 174, 26
Dianchi	1998–2007	none
Chao Lake	1998–2002	none
	2003–2007	262, 263, 267

Notes: 1. See the text for how to identify the regulated industries. 2. According to the official *Handbook* that describes major pollutants emitted by each 4-digit industry, industries in the following brackets are not classified as polluters in the associated 2-digit industry classification: 13 (131, 132, 1399), 22(223), 26 (2625, 2659, 2664, 2669).

basin, and 3) listed in the prevention-and-control plan for the basin. Table 2 lists the regulated industries in each basin and time period.

2.3. Developing and implementing regulatory measures

Targeted industries in a key county may be subject to a range of regulatory measures, which include command-and-control approaches (e.g., emission standards), economic incentives (e.g., emission charges), administrative measures (e.g., shutting down small or inefficient paper mills), and public disclosure. For example, in the 3Rs3Ls basins, paper mills with poor environmental performance, especially small ones, were forced to shut down to improve water quality in the adjacent rivers or lakes. According to the prevention-and-control plans, plants of specific industries within the key basins were required to reduce COD emissions at their own costs. The central government launched two environmental campaigns to improve enforcement of the regulations. The first one was called “shiwu xiao” (in English: fifteen small), which aimed to close fifteen types of small severely-polluting firms. During the first four years of implementing the 3Rs3Ls policy, more than 22,000 small polluting firms in the 3Rs3Ls basins were shut down (ECCEY, 2000, page 271). Meanwhile, larger paper mills were formed in the basins. The second campaign was called “shuang dabiao” (in English: meeting two standards). It aimed to make sure that larger polluting firms reach certain emission standards.

While local environmental authorities carry out most of the inspections, the central government conducts an annual review of environmental quality and enforcement efforts in the 3Rs3Ls basins. Plants located in a key county that emit COD are normally subject to relatively tougher regulatory oversight than are emitters in non-3Rs3Ls counties.

Many authors argue that China's existing regulations are weakly enforced or even ignored (see, e.g., van Rooij, 2006), so there are doubts about whether the existing regulations are, indeed, improving environmental quality in China. However, there is evidence that total COD emissions are decreasing at a faster rate in the 3Rs3Ls basins than in the non-3Rs3Ls areas. For example, the annual growth rates of COD emissions were almost the same for the 3Rs3Ls basins and the non-3Rs3Ls areas in the pre-regulation period, but were quite different in the post-regulation period (see Table 3).[9] This, together with our finding that the regulations had little effect on the emission levels of surviving firms (reported later in this paper), indicates that the initial implementation of the water quality regulations were mainly focused on small, heavily polluting firms, many of which were forced to shut down. As “low-hanging fruits” for COD reduction have been picked and as the implementation of the regional strategy for controlling COD is increasingly focused on large and surviving firms, it is important to understand how the regulations will affect these firms' productivity and emission levels.[10]

3. Data

3.1. Data sources

We construct data samples from three major sources. The first one is the Annual Surveys of Industrial Firms (ASIF) from 1998 to 2007, which were conducted by the National Bureau of Statistics of China (NBS). The annual survey covers all state-

[9] Some previous studies (see, for example, Vennemo et al., 2009) also find that industrial COD discharge fell significantly in China during the 2000s. Also, the surface water quality of major river basins improved slightly.

[10] There is a large body of literature on estimating the impacts of environmental regulations on emissions and/or environmental quality. Auffhammer and Kellogg (2011) examined whether US gasoline content regulations have successfully reduced ozone pollution by using daily measurements of ambient ozone concentrations from air quality monitors. Greenstone (2004) and Henderson (1996) used county-level data to examine the impact of a county's attainment status on local air quality.

Table 3
Industrial COD emissions in 3Rs3Ls and non-3Rs3Ls cities: Mean levels (in tonnes) and annual growth rates (%).

Regions	# cities	1992–1996			2004–2007		
		Emission, 1992	Emission, 1996	Annual growth	Emission, 2004	Emission, 2007	Annual growth
Non-3Rs3Ls	35	34,975	30,927	−3.03%	22,171	21,188	−1.13%
3Rs3Ls	11	47,183	41,740	−3.02%	32,503	29,362	−2.51%

Notes: 1. Data sources are various editions of the *China Environment Yearbook*. Data for 1994 and 2003 are not available. 2. The 1992–1996 period is regarded as pre-regulation; this is because 1996 is the first year during which the first counties in the 3Rs3Ls were identified as a key region for the water quality regulations. The 2004–2007 period is considered post-regulation period. That is, 2004 is the first year during which the 3Rs3Ls counties were fully determined. 3. The cities covered by the yearbooks vary across years. The mean levels and growth rates are calculated based on all cities where data for the selected years are available.

owned firms plus all non-state-owned firms with annual sales of more than 5 million RMB (about US $800,000). The ASIF data contain information about each firm's county of location, industry, birth year, ownership,[11] employment, fixed assets, gross output, wage payments, and other firm characteristics. Sectors in the ASIF include mining, manufacturing, and public utilities. The ASIF data have been employed by several recent studies on firm behavior, productivity, and economic growth in China (see, for example, Cai and Liu, 2009; Hsieh and Klenow, 2009; Brandt et al., 2012; Brandt et al., 2017).

In the ASIF, each firm is classified into an industry following the 4-digit Chinese Industry Classification (CIC) system. In 2003, the classification system was revised to incorporate more details for some sectors, while some other sectors were merged. Following Brandt et al. (2012), we construct a consistent classification so that the industry codes are comparable over the period. We also construct consistent area codes for counties in the data by using historical information on changes of county name and boundaries.[12] The present paper focuses on the manufacturing sector.

The second major data source for this analysis is the water quality regulation data that we compiled for the 3Rs3Ls basins. As discussed in section 2, we compiled the water quality regulation data using information from a number of sources.

The third major data source for this analysis is the Environmental Survey and Reporting (ESR) administrated by the Ministry of Environmental Protection (or the former State Environmental Protection Administration). China started collecting environmental statistics on industrial pollutants and waste in the 1980s. One of the main tasks for the ESR is to collect information about major polluting sources, including heavily polluting industrial enterprises, hospitals, residential pollutant discharging units, hazardous waste treatment centers, and urban sewage treatment plants. The ESR covered only about 3500 key industrial enterprises in the country during the 1980s. Since 1991, the scope of the ESR has been widened. Between 1991 and 1995, the survey covered all state-owned enterprises except those owned by a township.[13] From 1996 to 2000, township enterprises were also included in the ESR. Since 2001, the scope of the ESR has been determined by pollution discharges of individual firms/plants and the total amount of discharge in the county. Top firms/plants contributing 85% of total emissions of the major pollutants in a county are entered in the ERS. Thus, if a firm has two plants in two different counties, the reporting status of each plant depends on the plant's rank among polluting firms/plants in its county of location. Major pollutants considered in the ESR are COD, ammonia nitrogen, sulfur dioxide, industrial smoke and dust, and solid waste.

The ESR requires each surveyed firm/plant to complete a set of tables and submit them to the local (usually county level) environmental protection bureau (EPB). After receiving the finished tables, the local EPB checks the information and reports it to the EPB at a higher level. National key emitting enterprises designated by the Ministry of Environmental Protection report their survey tables every season. Beyond basic information such as name, industrial sector, location, and output value of the emitters, the ESR also includes emissions of major pollutants (e.g., COD) at the plant or firm level.

3.2. Sample construction and summary statistics

We merge the ASIF and the water quality regulation data by firm location and industry. We link the ESR data to the merged ASIF and water quality regulation data primarily by using firm legal codes. However, many observations in the ESR data have no firm legal code or have an incorrect legal code. We try to link such observations to the ASIF data using the name and location of the firms/plants. Both datasets report output value for the firms. This variable is very useful when we merge the

[11] Firms are legally registered as state-owned, collectively owned, private, domestic joint, foreign-invested, Hong Kong-, Macao-, and Taiwan-invested enterprise, or some others. A firm can be registered as a foreign-invested firm if at least a third of the ownership is foreign-held. Each firm in the survey reports its legal registration, which gives us information on ownership.

[12] The administrative divisions of China consist of five levels: central, provincial, prefectural, county, and township. There are three types of county-level administrative units in China: counties (in Chinese: *xian*), county level cities (in Chinese: *xian-ji-shi*), and wards of prefecture level cities (in Chinese: *shi-xia-qu*). Though a county or a ward of the four municipalities (Beijing, Tianjin, Shanghai, and Chongqing) directly under the central government has a higher status, it is treated as a county-level unit in our later analysis. Each county-level unit has a unique code which is published by National Bureau of Statistics. The code of a county will change in a later year if it becomes a ward of prefecture, splits into multiple counties, or combines with another county to form one single county. We recombine all splits to define counties with consistent boundaries over time. A consistent area code is used for a county. We also observe that some counties are incorrectly coded in the dataset for the firms. To correct the area codes and make them consistent over 1998–2007, we use standard codes published by NBS and information on firm's name, address, postal codes etc.

[13] In China, a state-owned enterprise usually has a rank in the administrative hierarchy of the country.

Table 4
The matched data sample: Mean levels of inputs and outputs.

Year	# firms	Gross output	Capital stock	Employment	Intermediates	COD emission	COD intensity
1998	3440	127.56	113.71	928	103.58	399.47	8.43
1999	4137	138.15	111.33	957	110.53	360.31	6.62
2000	4145	167.73	125.62	973	134.83	278.54	4.81
2001	4962	169.97	118.57	785	134.93	243.94	4.37
2002	4537	207.68	130.99	778	159.23	220.88	3.71
2003	4135	254.43	147.80	752	195.46	232.17	3.06
2004	4624	271.45	139.81	653	206.94	208.68	3.07
2005	4534	350.44	182.04	733	271.22	217.19	2.36
2006	3577	431.79	236.21	762	324.52	257.29	2.26
2007	4075	468.61	231.60	654	359.82	219.17	1.73

Note: Gross output and intermediates are measured in million 2007 RMB. COD in tonnes. COD intensity is defined as the COD emissions per unit of output value.

two firm-level datasets.[14] If the output values of a firm from the two datasets differ by more than 10%,[15] the firm will be excluded from the final matched sample. For the matched sample, both gross output and intermediates are measured in 2007 RMB to calculate growth rates between years. The obtained samples include firms from both the 3Rs3Ls basins and the non-3Rs3Ls region.

In this paper, we mainly focus on eight major COD-emitting manufacturing industries (CIC: 13, 14, 15, 17, 22, 26, 27, 32) targeted by the water quality regulations in the 3Rs3Ls basins. The matched data sample for these industries contains information from all the three data sources. We use this data sample to quantify the impact of the water quality regulations on firm's COD emissions and the effects of emission reduction on firm's output.

Table 4 provides a brief description of the matched data set for the eight major COD-emitting industries. We have about 4000 COD-emitting firms per year from 1998 to 2007. In contrast, the full ASIF sample has about 60,000 firms per year from the eight COD-emitting industries.[16] The average annual output per firm in our matched sample is 257 million RMB, which is about 3 times the average for firms in the eight industries in the ASIF data.[17] This suggests that a majority of small firms were not included in China's Environmental Survey and Reporting System. The matched data sample accounts for about 20% of gross output of the eight industries. The sampled firms emitted about 89 million tonnes of COD in 2007, which account for more than 17% of total emission from the whole industrial sectors. Table 4 also indicates that, over the years, COD intensity (i.e., COD emissions per unit of output value) has decreased significantly for the sampled firms.

A firm's COD intensity is correlated with its characteristics. Larger firms usually have a lower emission intensity. Compared to the firms from the central and western provinces, firms located in eastern China tend to have a lower COD emission intensity. Firms of different ownership types also tend to have different emission intensity levels. Specifically, foreign-invested and Hong Kong-, Macao-, and Taiwan-invested firms tend to have lower emission intensity than domestic firms. Among the three types of domestic firms, the collectively-owned enterprises tend to have better environmental performance.[18] The correlations are consistent with those reported by Wang and Jin (2007). In their study using Chinese city-level data, Cole et al. (2011) also found that Hong Kong-, Macao-, and Taiwan-invested firms have a moderate beneficial effect on water pollutants.

Table 5 presents mean annual growth rates for firms that exist in the sample for two consecutive years.[19] On average, real output of the firms grew by 3.5% annually from 1998 to 2007, while COD emission decreased by about 5.8% annually. This suggests a significant reduction in COD intensity for the sampled firms over the years.

4. Water quality regulations and firm productivity

Following Greenstone et al. (2012), we use a simple conceptual framework to guide our empirical investigation. Assume that a firm uses capital (K), labor (L), and materials or intermediates (M) to produce desired output (Y). The firm's potential

[14] As discussed earlier, an observation in the ESR may represent a plant of a firm. In contrast, the ASIF data are firm-level information. If a firm contains plants in multiple counties, the firm will appear as one observation in the ASIF data.

[15] We do the comparisons using current price from both datasets. Later we will check the robustness by using a stricter criterion of matching the data. See section 7.5 for details.

[16] Over 90% of ASIF firms are dropped when obtaining the matched data sample. There are three major reasons: First, the ESR only contains major polluting firms/plants in each county. Those smaller COD-emitting manufacturing firms are not included. Second, the level of observation in the ESR is firm/plant in a county. However, the observation in the ASIF data is firm which may consist of multiple plants in different counties. If not all plants of a firm are included in the ESR, we are unable to link the two datasets. Third, there are recording errors in the surveys so that the output value of a firm might be different in the ESR and ASIF for some firms. In that case, the firm is deleted from the sample because we can't determine which survey reports the true output value.

[17] See Table A3 in the appendix for the mean levels of inputs and output for the eight COD-emitting industries in the ASIF data.

[18] See Table A4 for the regression.

[19] To address the concerns of outliers, we delete 1% of the observations with extreme values of growth rates in employment, capital stock, intermediates, or output value.

Table 5
The one-year difference data sample: Mean growth rates of inputs and outputs.

Period	# firms	Gross output	Capital stock	Employment	Intermediates	COD emissions
1998–1999	1377	−0.005	0.004	−0.048	−0.018	−0.110
1999–2000	1633	0.044	0.003	−0.041	0.049	−0.097
2000–2001	1678	0.007	0.011	−0.037	−0.002	−0.057
2001–2002	2131	0.042	0.015	−0.035	0.039	−0.056
2002–2003	1848	0.045	0.041	−0.024	0.057	−0.059
2003–2004	1617	0.034	0.040	−0.030	0.033	−0.049
2004–2005	1895	0.051	0.033	−0.009	0.034	−0.047
2005–2006	1509	0.062	0.041	−0.009	0.040	−0.058
2006–2007	1590	0.073	0.045	−0.003	0.074	−0.041

Note: Growth rate is defined as $\Delta\log(x)$.

output, Y^p, is given by:

$$Y^p = Af(K,\ L,\ M),$$

where A is a Hicks-neutral technology shifter. The production generates emissions. In order to comply with environmental regulations, the firm makes efforts to limit emissions. The final output, Y, is represented by

$$Y = (1-\theta)Af(K,\ L,\ M),$$

where θ is the fraction of the production effort used for limiting emissions and equals zero when there is no environmental regulation. The firm's measured TFP is then:

$$TFP = (1-\theta)A,$$

which shows that environmental regulation that forces firms to allocate resource to pollution control will reduce their TFP. When facing new environmental regulations or pollution reduction requirements, firms may choose to invest in "changes in process" technologies, "end-of-pipe" technologies, or do both (Berman and Bui, 2001). Thus, environmental regulation may affect productivity through its effect on firms' efforts to limit emissions under the current technology or to invest in cleaner technologies.

We estimate various versions of the following model to measure the effect of the water quality regulations on firms' productivity:

$$TFP_{fjct} = \omega I\left(COD_{ft} \times 3Rs3Ls_{ct}\right) + \beta X_{ft} + \eta_{ct} + \delta_{jt} + \mu_f + u_{fjct}, \tag{1}$$

where TFP_{fjct} denote total factor productivity for firm f of industry j in county c at year t. The indicator function $I(\cdot)$ defines the regulatory status of a firm. It equals one if and only if firm f is a regulated COD emitter (i.e., $COD_{ft} = 1$) located in a 3Rs3Ls county (i.e., $3Rs3Ls_{ct} = 1$). X_{ft} is a vector of firm characteristics. It includes dummy variables indicating firm ownership types: state-owned enterprise; collectively-owned enterprise; private enterprise; Hong Kong-, Macao-, and Taiwan-invested enterprise; foreign-invested enterprise; and the domestic joint venture.[20] X_{ft} also includes firm size indicators based on gross output (i.e., smaller than the median; between the median and the 75% of mean; between the 75% of mean and the mean; and greater than the mean in the industry-year group); the firm age indicator (which is equal to one if the firm is 10 years old or older); the ratio of export value to sales; a dummy indicating whether the firm has multiple plants; and the total employment of the firm's 2-digit industry in the same county. These factors were found to be correlated with firm growth in the US (Greenstone, 2002), China (Brandt et al., 2017; Wang and You, 2012), and other developing countries (Dethier et al., 2011).[21]

Equation (1) also controls for a) local economic shocks with county by year fixed effects (η_{ct}), b) annual fluctuations by industry with 4-digit industry by year fixed effects (δ_{jt}), and c) firm fixed effects (μ_f). The last one, u_{fjct}, is an error term. The identification assumption of model (1) is that unobserved factors are uncorrelated with the regulation indicator, conditional on the covariates: $E[I(\cdot)^* u_{fjct} | X_{ft}, \eta_{ct}, \delta_{jt}, \mu_f] = 0$. The key parameter is ω, which represents the effect of the water quality regulations on TFP of regulated firms in the 3Rs3Ls basins.

A firm's TFP is calculated by

[20] Since the late 1990s, China introduced various measures to dramatically reform its state-owned firms (Hsieh and Song, 2015). To control for ownership-by-year impacts, the regressions allow the effects of the (six) ownership categories to vary across years.

[21] Greenstone et al. (2012) estimated models without these characteristics. We also tried that and found very similar results as reported later in this section. The results are available from the authors upon request.

Table 6
Effects of the water quality regulations on total factor productivity.

	(1)	(2)	(3)	(4)	(5)
Water quality regulation	0.0012	−0.0008	0.0008	−0.0013	−0.0061
	(0.0028)	(0.0017)	(0.0017)	(0.0020)	(0.0068)
Water quality regulation*log(COD intensity)					−0.0014
					(0.0018)
Firm and county characteristics	Y	Y	Y	Y	Y
4-digit industry by year FE	Y	Y	Y	Y	Y
County FE		Y			
County by year FE			Y	Y	Y
Firm FE				Y	Y
N	1,937,283	1,937,269	1,936,393	1,795,698	1,795,698

Notes: 1. Results are obtained from regressing various versions of equation (1). 2. Robust standard errors clustered at the county level are reported in parentheses.

$$TFP_{ft} = y_{ft} - \alpha_{lt} l_{ft} - \alpha_{kt} k_{ft} - \alpha_{mt} m_{ft},$$

where y_{ft} is logged output for firm f in year t. Logged labor, capital, and intermediate inputs are represented by l_{ft}, k_{ft}, and m_{ft}, respectively. Here, α_{lt} is labor's share of gross output which takes the same value for all firms from the same 4-digit industry in year t. Several recent studies of Chinese manufacturing firms' productivity (see, for example, Hsieh and Klenow, 2009; Brandt et al., 2012) used this approach to calculate TFP. Greenstone et al. (2012) used a similar method to calculate firms' TFP in an analysis of the impact of the air quality regulations in the US. We follow Brandt et al. (2012) to adjust labor share (i.e., inflate wage payments to 50% of value-added for each 4-digit industry) and labor input (i.e., using wage payments as the measure to capture increased hours worked and accumulated human capital content) and to construct data series of gross output and intermediate inputs in constant RMB.

For the empirical analysis reported in this section, we use all manufacturing firms from the ASIF data and the water quality regulation information. Following the common practice for using the ASIF data (Cai and Liu, 2009; Hsieh and Klenow, 2009), we drop outliers from the original sample. A firm is considered an outlier if its calculated TFP is larger than the 99th percentile or smaller than the 1st percentile in each industry-year group.

Table 6 reports the estimated effect of the water quality regulations on firms' productivity in the 3Rs3Ls basins. Column 1 controls for firm and county characteristics and the fixed effects of 4-digit industry by year. The estimates suggest that the water quality regulations in the 3Rs3Ls basins had no statistically significant impact on firms' TFP. Column 2 additionally includes county fixed effects, and again the results indicate that the water quality regulations had no statistically significant effect on firms' TFP. Controlling for industry by year and county by year fixed effects also does not change the results, as shown in column 3. Column 4 additionally controls for firm fixed effects, so the influence of all time-invariant firm-level factors that affect TFP is removed. Though the estimated coefficient indicates a negative impact of the water quality regulations, it is again statistically insignificant at conventional levels.[22]

An industry that has higher COD emission intensity may face stricter regulation. In the last column of Table 6, we add the product of the regulation dummy and the logged value of industrial COD intensity in 1995, a year before the 3Rs3Ls policy was implemented (ECCEY, 1996). This measure is similar to the one used by Hering and Poncet (2014) to estimate the impacts of SO_2 regulation in China. Again, the results suggest that the 3Rs3Ls policy has no statistically significant impacts on a firm's TFP.[23]

As discussed in the introduction, there are two possible explanations for the insignificant effect of the water quality regulations on firm productivity. One is that the regulations were not well enforced and therefore they did not affect individual firms' productivity. The second explanation is that environmental regulation does not necessarily reduce firms' productivity, as suggested by the Porter hypothesis. In the next two sections, we test which explanation is more plausible.

5. Are the water quality regulations ineffective?

The most direct explanation for the insignificant effect of the water quality regulations on TFP is that the regulations are ineffective in forcing the firms to reduce their emissions and, therefore, do not affect their TFP. To test this hypothesis, we estimate an emission function that links a firm's COD emission level to its water quality regulation status.

[22] We used a threshold to define heavy emitters. In this paper, we mainly report the results that correspond to a threshold of 3.5%. The results are robust if we use the central government's list of targeted industries as the regulated COD-emitting industries. The results are available upon request from the authors.

[23] We also did robustness checks and heterogeneity analyses for the TFP impacts of the 3Rs3Ls policy. In general, the checks and analyses confirm the results. As the paper focuses on the explanations for the TFP results and providing more evidence about the impacts of regulation on emissions and the effects of emission abatement on productivity, we do not report the checks and analyses here due to space limit. However, the results are available from the authors upon request.

To model the effects of regulation on COD emissions, we follow Copeland and Taylor (2003) and write the emission function as

$$Z = \phi(\theta)Z^p,$$

where Z^p is the emission level without any abatement by the firm. Again, θ is the fraction of the firm's production effort devoted to emission control. By choices of units, potential emissions can be written as

$$Z^p = Y^p.$$

The regulation affects actual emissions by forcing the firm to change θ. The following equation links actual COD emissions to production inputs and abatement efforts:

$$Z = \phi(\theta)Af(K, L, M).$$

This equation suggests that changes in emission levels can be related to changes in firm's production effort devoted to emission abatement. Assuming a Cobb-Douglas form for $f(\cdot)$, we test the first explanation regarding the TFP impact of the water quality regulations by estimating the following equation:

$$\Delta\log Z_{fjct} = \omega_1 I\left(COD_{ft} \times 3Rs3Ls_{ct}\right) + \delta_1\Delta\log K_{ft} + \gamma_1\Delta\log L_{ft} + \sigma_1\Delta\log M_{ft} + \mu_1 X_{ft-1} + \eta_{ct} + \delta_{jt} + \mu_f + \varepsilon_{fjct}, \quad (2)$$

where $\Delta\log Z_{fjct}$ is emission growth rate from year $t-1$ to year t for firm f of 4-digit industry j in county c, and X_{ft-1} is a vector of firm and county characteristics in year $t-1$ including COD intensity of the firm. This model also controls for a set of fixed effects: local economic shocks with county by year fixed effects (η_{ct}), temporal industrial fluctuations with industry by year fixed effects (δ_{jt}), and firm fixed effects (μ_f). ε_{fjct} is an error term. The time-differencing model (2) should help address the estimation issue caused by omitted variables in the emission function.

The equations assume that the firm generates emissions as a result of using specific materials or intermediates (e.g., coal and water). The emission volume would be affected by the firm's characteristics such as age and type of ownership which may affect the technology adopted (Cole et al., 2011; Wang and Jin, 2007). Exporting status and international linkage may also affect a firm's environmental performance (Lin et al., 2014). To control these firm characteristics, the regression includes log(1 + firm age) and the squared and the ratio of export value to sales. Furthermore, the regression controls scale economies in limiting emissions by including a firm size variable (logged output value). The regression also controls for knowledge spillovers by including a variable for logged total employment of the firm's 2-digit industry in the same county.

Table 7 presents five sets of results from estimating model (2). The first four sets of results are obtained by using a firm's COD regulatory status (i.e., if it belongs to a regulated industry in the 3Rs3Ls basins). All of these results suggest that the water quality regulations in the 3Rs3Ls basins had no statistically significant effect on firms' emission levels. The estimated coefficients can be thought of as the mean effect of the water quality regulations on firms' COD emissions over years.

When developing and enforcing the water quality regulations, the government may focus on firms with high emission intensity. Therefore, firms with higher pollution intensity may be subject to more stringent regulation and enforcement efforts. To explore this possibility, we include an interaction term between the firm's COD regulatory status and its COD emission intensity. The COD emission intensity is likely endogenous. Therefore, we use the products of logged value of precipitation and each of the eight 2-digit industry dummies as instrument variables for the interaction term: water quality regulation $*$ log(COD intensity). Precipitation[24] does not directly affect an industrial firm's COD emission, but has been found to affect governments' enforcement efforts of environmental regulation in China. Precipitation dilutes pollution concentrations in rivers and lakes and thus reduces the pressure for pollution control (Lin, 2013) and therefore should serve as a good instrument. Indeed, test results from first-stage regressions indicate that the instruments are valid: Kleibergen-Paap rk Wald F-statistic = 5.66, suggesting that the null hypothesis that the equation is weakly identified can be rejected at the 1% significance level; Hansen J-statistic = 8.15, suggesting that we can't reject the null hypothesis that the instruments are uncorrelated with the error term. The statistics imply that the instruments are exogenous and relevant. Column 5 of Table 7 reports the IV estimates. Again, the result shows that the water quality regulations in the 3Rs3Ls basins had no statistically significant effect on surviving firms' COD emissions.[25]

[24] The precipitation data come from the China Meteorological Data Sharing Service System, which reports daily minimum and maximum temperatures, precipitation, and many other weather attributes at 820 weather stations in China since the 1950s. See Chen et al. (2016) for more details.

[25] When we use the interaction term [water quality regulations*log(COD intensity)] for the first 3 columns of Table 7, we obtain qualitatively the same results regarding the impacts of the water quality regulations on firm's emission. See Table A5 in the appendix for results.

Table 7
Effects of the water quality regulations on COD emissions.

	Dep. Variable: Δlog (COD emission)				
	(1)	(2)	(3)	(4)	(5)
Water quality regulation effect	0.0264	−0.0013	0.0136	−0.1123	−0.1348
	(0.0263)	(0.0347)	(0.0466)	(0.1198)	(0.1103)
Water quality regulation* *log*(COD intensity)					−0.2265
					(0.1683)
Firm characteristics	Y	Y	Y	Y	Y
4-digit industry by year FE	Y	Y	Y	Y	Y
County FE		Y			
County by year FE			Y	Y	Y
Firm FE				Y	Y
N	15,193	14,808	11,213	7461	7461

Notes: 1. Results are obtained from regressing various versions of equation (2). 2. ***, **, and * denote significant at 1%, 5%, and 10%, respectively. Robust standard errors clustered at the county level are reported in parentheses. 3. See the text for model specifications and estimation methods for the five columns. Column 5 reports IV estimates.

6. Is Porter right?

The second possible explanation for the insignificant effect of the water quality regulations on TFP is that limiting emission does not necessarily reduce firms' productivity, as suggested by the Porter hypothesis. To explore this explanation, we estimate a production function that models emission as an input for producing the desired output (Cropper and Oates, 1992). This equation provides evidence on the validity of the second explanation – if COD emissions are an inevitable by-product of producing desirable goods, then efforts to reduce COD emissions will necessarily reduce firms' output under the current production technology.

Earlier, the production function for Y is represented by

$$Y = (1-\theta)Af(K,\ L,\ M).$$

Although it is difficult to measure a firm's emission control effort, θ, the outcome of the effort, changes in emission level, is more visible. Following Copeland and Taylor (2003), the emission function is assumed to be

$$Z = \phi(\theta)Af(K, L, M),$$

where Z is the emission level observed. As Copeland and Taylor (2003) showed, the specification for Z here can be interpreted as being supported by a pollution abatement technology. With the assumption that $\phi(\theta) = (1-\theta)^{1/\rho}$, the observed output is then given by

$$Y = Z^{\rho}[Af(K, L, M)]^{1-\rho}.$$

A direct OLS estimation of the above equation may suffer from the omitted variable problem widely known in the production economics literature. One can try to control for unobservables using the approaches proposed by Olley and Pakes (1996) and Levinsohn and Petrin (2003), but the appropriateness of these approaches depends on a number of assumptions, including a perfectly competitive market (Griliches and Mairesse, 1995). Some of the assumptions may not hold for China's polluting firms.[26]

Following previous studies on estimating production functions (e.g., Haskel et al., 2007), we estimate a time-differencing model which takes the following form

$$\Delta\log Y_{fjct} = \rho_2\Delta\log Z_{ft} + \delta_2\Delta\log K_{ft} + \gamma_2\Delta\log L_{ft} + \sigma_2\Delta\log M_{ft} + \mu_1 X_{ft-1} + \eta_{ct} + \delta_{jt} + \mu_f + \varepsilon_{fjct}, \tag{3}$$

where $\Delta\log Y_{fjct}$ is output growth rate between year t-1 and year t for firm f of 4-digit industry j in county c, X_{ft-1} is a vector of firm characteristics in year t-1,[27] and ε_{fjct} is an error term. This model also controls for a set of fixed effects: local economic shocks with county by year fixed effects (η_{ct}), temporal industrial fluctuations with industry by year fixed effects (δ_{jt}), and firm fixed effects (μ_f). The time-differencing equation controls for time-invariant unobservables that may affect output. We are particularly interested in ρ_2, which denotes output elasticity of COD emissions. This elasticity estimate measures the opportunity cost for reducing COD emissions (in terms of loss in output value) under the current production technology.

[26] As a robustness check, estimates from fixed-effect models and a model proposed by Levinsohn and Petrin (2003) will be presented later in Section 7.
[27] See the previous section for the variables other than COD intensity.

Table 8
Effects of COD emission reductions on output value.

	Dep. Variable: Δlog (output)					
	(1)	(2)	(3)	(4)	(5)	(6)
Δlog (COD emission)	0.0092***	0.0090***	0.0103***	0.0117***	0.0118***	0.0122***
	(0.0014)	(0.0015)	(0.0020)	(0.0029)	(0.0029)	(0.0031)
Water quality regulation					0.0223	0.0222
					(0.0220)	(0.0220)
Δlog (COD emission)*Water quality regulation						−0.0033
						(0.0089)
Firm characteristics	Y	Y	Y	Y	Y	Y
4-digit industry by year FE	Y	Y	Y	Y	Y	Y
County FE		Y				
County by year FE			Y	Y	Y	Y
Firm FE				Y	Y	Y
N	15,193	14,808	11,213	7461	7461	7461

Notes: 1. Results presented in the table are obtained from regressing various versions of equation (3). 2. ***, **, and * denote significant at 1%, 5%, and 10%, respectively. Robust standard errors clustered at the county level are reported in parentheses. 3. Due to space limitations, we do not report estimated coefficients for firm characteristics which are available upon request from the authors. A general pattern is that young, large foreign invested enterprises produce more output for a given level of input.

Table 8 reports regression results from six alternative specifications of model (3). Specification 1 controls for firm and county characteristics. Changes in logged values of labor, capital, and intermediate inputs are also included in the regression. To control for shocks to the associated 4-digit industry in a year, the regression includes 4-digit industry by year fixed effects. The column presents an estimate of output elasticity of COD emissions, i.e., the percent change in output value when COD emissions are reduced by one percent. The result suggests that limiting COD emissions has a negative impact on output value. A 1% reduction in COD emissions is associated with about 0.01% loss in total output value of the desired output. Specification 2 additionally includes county fixed effects in the regression, and the results barely change. In specification 3, we add county by year fixed effects to control for annual shocks to the local economy. Specification 4 additionally controls for firm fixed effects and is our preferred specification because it presents estimates of the impacts that are purged of all firm-level growth determinants and transitory determinants that are common to all firms within the same county or industry. The results suggest that reducing COD emissions by 1%, whether it is voluntarily or compulsorily, would decrease output value by about 0.012%. This estimate is statistically significant at the 1% level. The estimated coefficient is in line with results reported by related literature. Using data from the manufacturing industry in Western Germany to fit a Cobb-Douglas production function, Keilbach (1995) found that output elasticities of emissions of SO_2, CO_2, NO_x, and particulate matter range between 0.01 and 0.07.

Environmental regulation may stimulate innovation and technological change. Previous studies have provided evidence of such effects in terms of R&D spending (Jaffe and Palmer, 1997; Lanoie et al., 2011), patents (Brunnermeier and Cohen, 2003; Calel and Dechezleprêtre, 2016; Popp, 2006), and productivity (Yang et al., 2012; Lanoie et al., 2008; Rexhäuser and Rammer, 2014). Others, however, have found that environmental regulation has no or even negative effects on innovation and productivity (see Ambec et al. (2013) for a comprehensive review). For our case of water quality regulations in China, specification 5 of Table 8 assumes a Hicks-neutral technological change and estimates a corresponding equation.[28] The estimated results do not provide supportive evidence of technological progress stimulated by the water quality regulations in the short run. In the last column of Table 8, we include the interaction term between the change in COD emissions and the water quality regulation dummy. The result indicates that the water quality regulations do not change the output elasticity of the pollutant "input." A comparison among the six sets of results shows that the estimated output impacts of COD emissions are quite similar across different specifications. In the next section, we offer further evidence from heterogeneity analysis and robustness checks.

7. Heterogeneity analysis and robustness checks

Our empirical analysis suggests that the water quality regulations in the 3Rs3Ls basin had no statistically significant effect on firms' productivity because the regulations were ineffective in reducing surviving firms' COD emissions. An effective regulation that forces firms to reduce their COD emissions would reduce their output values. In this section, we check the robustness of these results and demonstrate that they are quite stable across regions, industries, firm sizes, and ownerships.

7.1. Do the effects vary across regions?

There may be reasons to suspect that the effects of the water quality regulations vary by regions. Because different regions may be at different levels of economic development, they may put different priorities on environmental protection and,

[28] That is, adding the regulatory measure to the right-hand side of equation (3).

Table 9
Effects of the water quality regulations on COD emissions and effects of COD emission reductions on output value, by region.

	Eastern	Central and Western
	(1)	(2)
Panel A: Dep. Variable: Δ*log*(COD emissions)		
Water quality regulation effect	−0.1023	−0.1957
	(0.1294)	(0.3050)
Panel B: Dep. Variable: Δ*log*(output)		
Δ*log*(COD emissions)	0.0083**	0.0301***
	(0.0032)	(0.0065)
#obs.	6179	1282

Notes: 1. See Fig. 1 for the boundaries of the regions. 2. Estimates in Panel A are obtained from regressing equation (2), which includes interactions of the region categories and regulatory status of firms in addition to other control variables used in column 4 of Table 7. Panel B results are obtained from regressing equation (3), which includes interactions of the region categories and Δlog (COD emission) and controls for variables used in column 4 of Table 8. 3. The last row reports the number of firm observations from each region.

therefore, may devote different amounts of effort to enforce the 3Rs3Ls regulations. The counties in our sample can be divided into the eastern, central, and western regions according to the central government's classification. The majority of the sampled firms are located in the eastern region.

To check if the effects of the water quality regulations vary across regions, we re-estimate model (2) by including interaction terms of the water quality regulations measure and the regional dummies for the eastern and the other (i.e. central and western) regions.[29] The results are reported in Panel A of Table 9. Both coefficients for the two regions are negative, but are statistically insignificant. Overall, the findings are consistent with those reported earlier that the 3Rs3Ls policy did not significantly reduce firms' COD emissions.

We also re-estimate model (3) by allowing regional differences in the opportunity cost of emission reduction. The results are reported in Panel B of Table 9. Again, we find that firms cannot reduce their emissions without sacrificing some output values under the current production technology. The effect is smaller in the eastern region. Our data do not allow us to probe the reasons for the regional differences. One possible explanation is that firms in the central and western regions have lower capacity to abate COD because they tend to use less advanced production technologies.

7.2. Do the effects vary across industries?

There may also be reasons to suspect that the effects of the water quality regulations vary by industry because different industries tend to use different production technologies. In addition, the stringency of the regulations may vary across industries. The top panel of Table 10 presents results from a regression model that allows the impacts of the regulations on COD emissions to vary across industries. As some industries have a small number of regulated firms in the sample,[30] we consider some 2-digit industries as a single group. The estimates indicate that the water quality regulations had no statistically significant effects on emissions for any of the five industrial groups.

To test if the opportunity cost of emission reduction varies across industries, we modify model (3) to allow the cost elasticity to vary across industrial groups (see Panel B of Table 10). The regression results indicate that the opportunity cost for COD emission reductions vary across industrial groups. The estimated coefficients for four out of the five industries take positive values. For the three industry groups which have the largest number of observations, a 1% decrease in COD emissions would reduce output value by 0.013%–0.021%.

7.3. Do the effects vary by firm size?

Environmental regulation may also affect small and large firms differently because they may use different production technologies. Large firms may have lower marginal abatement costs (MAC) than small firms because of scale economies in pollution abatement (Dasgupta et al., 1997, 2001). Therefore, large firms tend to have better environmental compliance records than smaller firms (Becker et al., 2013). Using UK data, Cole et al. (2005) found that pollution intensity is negatively correlated to the size of the average firm in an industry.

Panel A of Table 11 presents estimates from a regression that includes a variable for water quality regulations and its interaction with firm size. The results indicate that the water quality regulations had a smaller impact on smaller firms' COD emissions than on larger firms' COD emissions, but the estimated coefficient on the interaction term is also statistically insignificant. Overall, the panel suggests that the regulations have no significant effects on emissions.

[29] As there are only about 500 observations for the western region in the regression model, we combine firms from the western and central regions into a single group.

[30] For instance, among the 7461 observations, there are only two regulated firms from the CIC13 industries in the sample. The number of regulated CIC32 firms is 6.

Table 10
Effects of the water quality regulations on COD emissions and effects of COD emission reductions on output value, by industry.

	CIC13, CIC14, and CIC15	CIC17	CIC22	CIC26	CIC27 and CIC32
	(1)	(2)	(3)	(4)	(5)
Panel A: Dep. Variable: Δ*log*(COD emissions)					
Water quality regulation	0.5247	−0.1391	−0.2487	−0.0573	0.2270
	(0.4321)	(0.1827)	(0.2527)	(0.185)	(0.3142)
Panel B: Dep. Variable: Δ*log*(output)					
Δ*log*(COD emissions)	0.0214***	0.0127**	0.0068	0.0146***	−0.0063
	(0.0078)	(0.055)	(0.0086)	(0.0049)	(0.0078)
#obs.	1807	1861	702	1967	1124

Note: 1. CIC13 = Processing of Food from Agricultural Products; CIC14 = Food; CIC15 = Beverages; CIC17 = Textile; CIC22 = Paper and Paper Products; CIC26 = Raw Chemical Materials and Chemical Products; CIC27 = Medicines; CIC32 = Smelting and Pressing of Ferrous Metals. 2. As some industries have a small number of regulated firms in the sample, we consider group "food-related" industries (CIC13, 14, and 15) into a single grand group. CIC27 and 32 are treated as a single for the analysis presented in the Table 3. The regression model controls for the shocks to the associated 4-digit industry in a year that are common to both regulated and unregulated firms.

Panel B presents estimates of the opportunity cost of emission reductions (in terms of loss in output value) with the assumption that it varies across firm sizes. Reducing COD emissions has larger marginal costs for smaller firms. This difference in opportunity cost of COD reductions may be due to the economy of scale in pollution prevention and control.

7.4. Do the effects vary by ownership?

Ownership may also affect how firms respond to environmental regulation. Pargal and Wheeler (1996) found that publicly-owned factories have higher marginal abatement costs than privately-owned factories. Panel A of Table 12 presents estimates of the impacts of the water quality regulations on COD emissions by ownership type. The results suggest that the regulations had no statistically significant effect on COD emissions for any ownership type. Panel B presents estimates of cost elasticity of COD emission reductions by ownership type. The cost elasticity is found to be positive for five out of the six types of firms. It is statistically different from zero for state-owned enterprises, domestic joint ventures, and foreign-invested enterprises.

Several studies have compared firms' environmental performance by ownership type and found that multinational enterprises (MNEs) tend to have cleaner technology. Many developed countries usually have higher environmental standards, which may induce innovation and development of environment-friendly technology (Lanjouw and Mody, 1996). Therefore, foreign-invested enterprises often employ newer, cleaner technology even in locations where standards are relatively weak. Domestic firms in many developing countries do not have the financial resources to acquire environmentally-friendly technologies, especially when faced with new entrants and foreign competition (Christmann and Taylor, 2001). MNEs usually face greater pressure for environmental protection. Institutional pressures for MNEs' environmental self-regulation originate in a complex legitimating environment, including supranational institutional pressures (Kostova and Zaheer, 1999). Customers and the general public may have much less tolerance with wrongdoings of foreign firms than with those of domestic firms (Lin et al., 2014). Foreign firms may also have less bargaining power than domestic firms (Lin et al., 2014). When the local government has other priorities (in most cases, economic growth), compliance with related environmental

Table 11
Effects of the water quality regulations on COD emissions and effects of COD emission reductions on output value: interactions with firm size.

	(1)	(2)
Panel A: Dep. Variable: Δ*log*(COD emissions)		
Water quality regulation	−0.1589	−0.7241
	(0.2681)	(0.6933)
Water quality regulation*log(output)	0.0153	0.0537
	(0.0232)	(0.0590)
Panel B: Dep. Variable: Δ*log*(output)		
Δ*log*(COD emissions)	0.0426**	0.0756***
	(0.0171)	(0.0243)
Δ*log*(COD emissions)*log(output)	−0.0029*	−0.0057***
	(0.0015)	(0.0021)
Firm characteristics	Y	Y
4-digit industry by year FE	Y	Y
County by year FE	Y	Y
Firm FE		Y
#obs.	11,213	7461

Table 12
Effects of the water quality regulations on COD emissions and effects of COD emission reductions on output value, by ownership.

	SOE	COE	DJV	Private	Foreign	HMTIE
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Dep. Variable: Δ*log*(COD emissions)						
Water quality regulation effect	0.1390	0.0247	−0.2479	0.0273	−0.0240	−0.2292
	(0.1794)	(0.2976)	(0.1558)	(0.2332)	(0.1990)	(0.3867)
Panel B: Dep. Variable: Δ*log*(output)						
Δ*log*(COD emissions)	0.0169***	−0.0099	0.0165***	0.0140	0.0186*	0.0094
	(0.0054)	(0.0090)	(0.0050)	(0.0101)	(0.0096)	(0.0096)
#obs.	1637	472	2510	757	1196	889

Note: SOE = state-owned enterprise; COE = collectively-owned enterprise; DJV = domestic joint venture; Private = private enterprise; Foreign = foreign-invested enterprise; HMTIE=Hong Kong-, Macao-, and Taiwan-invested enterprise.

regulations is often subject to negotiation (Child and Tsai, 2005). Wang and Wheeler (2003) found that firms with different ownership structures have different bargaining power concerning the enforcement of environmental regulations such as pollution charges and fines. Unfamiliar with the local political context, foreign firms are often targeted for regulatory enforcement.

We also check whether the regulation impacts vary across time periods. This is used to address the concern that the regulated firms are subject to changing regulation over time. In the check, we divide the ten-year period into two sub-periods and use 1998–2002 as the first period. The main reason for this divide is that most of the environmental plans for the 10th Five-Year Period was approved in 2003. Regression results indicate that the regulations had no statistically significant effect on COD emissions in both periods.

7.5. Robustness checks

An examination of data reveals that growth rates of COD emissions were different before and after 2001 (see Table 5). Moreover, in 2001, the environmental survey started to use a new and clearer rule to determine which polluting firms/plants are included in the survey.[31] Could this affect our estimation results? Column 1 of Table 13 presents of estimates of equations (2) and (3) using the data after 2001. The results again suggest that the water quality regulations had no statistically significant impacts on COD emissions in the 3Rs3Ls basins, and reducing COD emissions would reduce firms' output values under the current production technologies.

As discussed in the data section, the ASIF and the ESR may report different output values for a firm. A firm was excluded from our sample if the reported output values differ by more than 10%. Column 2 of Table 13 reports results estimated using a sample that uses 5% as the exclusion criterion. The results again indicate that the water quality regulations had no statistically significant effect on surviving firms' emissions, and it would be costly for firms to reduce their emissions under the current production technologies.

So far, we have estimated models (2) and (3) using the one-year difference of the data. We can also estimate these models using the two-year difference. The advantage of using one-year difference of the data is that more observations are available for the one-year difference than for the two-year difference. However, using longer time difference can attenuate the problem of measurement errors associated with differencing (Griliches and Hausman, 1986; Haskel et al., 2007). Column 3 reports results that are estimated using the two-year difference of the data. The results again show that the water quality regulations had no statistically significant effect on surviving firms' emissions, and it would be costly for firms to reduce their COD emissions under the current production technologies.

A firm could transfer production activities from a plant in a regulated area to a plant in a non-regulated area. Such potential displacements could bias the estimated regulation effect (in absolute value) downward. As a robustness check, we limit our data sample to firms with only one production unit. The results are reported in column 4 of Table 13. Again, the results suggest that the regulations had an insignificant impact on COD emissions. For the one-plant firms, a 1% reduction in COD emissions would cause output value to decrease by 0.009%. In another related check, a firm would be removed from the 1-year difference sample if it changed their county of location during two consecutive years. This makes the data sample decrease by only 521. Regression results again suggest insignificant impacts of the regulations on COD emissions and negative impacts of COD emissions on output value.

Surviving firms may have different production technologies than firms that entered or exited the market during the study period. As a robustness check, we exclude the firms that entered or exited the market during the study period and focus exclusively on firms that survived through the entire study period 1998–2007. This leaves us with about 1100 observations.[32]

[31] See section 3 for more information on the survey.

[32] To obtain the sample, we limit it to firms which were born on or before 1998 (i.e. the first year of study period) and appeared in our data in year 2007. Some of those firms may not appear in the sample in some years between 1998 and 2007 because of the ASIF survey criteria and/or matching the ASIF data and ESR data (see the data section for details). This regression does not control firm fixed effects. Otherwise, the number of observations decreases sharply.

Table 13
Robustness checks.

	Data from 2001	Matching data (5%)	Two-year difference	One-plant Firms	Survived 1998 –07	Regu. var. (10%)	Change in productivity	FE Model
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Dep. Variable: Δ*log*(COD emissions)								
Water quality regulation	−0.0921	0.1044	0.0021	−0.0948	0.0181	−0.1557	NA	−0.0789
effect	(0.1492)	(0.2201)	(0.1189)	(0.1294)	(0.1446)	(0.1291)		(0.0776)
Panel B: Dep. Variable: Δ*log*(output)								
Δ*log*(COD emissions)	0.0138***	0.0224***	0.0158***	0.0092***	0.0183**		0.0066**	0.0110***
	(0.0040)	(0.0047)	(0.0042)	(0.0035)	(0.0084)		(0.0031)	(0.0018)
#obs.	4811	3155	4355	6145	1107	7461	7461	25,869

The results reported in column 5 of Table 13 again indicate that the regulations had a statistically insignificant effect on COD emissions, and that it would be costly for firms to reduce their emissions under the current production technologies.

In the previous subsections, we assumed that the targeted industries for COD regulations are those that are listed in the prevention-and-control plans and emit 3.5% or more of total industrial COD emissions. As a robustness check, we redefine the regulated industries as the listed industries that emit 10%, instead of 3.5%, or more of total industrial COD emissions. Again, we obtain similar results (column 6).

We use changes in output value to measure the opportunity cost for COD emission reductions in the analysis above. Would the conclusion change if we use changes in calculated TFP to measure the opportunity cost? Results in column 7 provide the answer to this question.[33] Specifically, the results indicate that a 1% reduction in COD emissions will lead to a 0.007% reduction in the calculated TFP. This estimate is qualitatively the same as the baseline result.

In the previous regressions, we use the one-year difference data, which have many advantages as discussed earlier. However, the constructed sample excludes many observations as many firms did not exist for two consecutive years. In the last column of Table 13, we report results from a fixed-effect model that uses output value as the dependent variable.[34] Again we find qualitatively the same results. We also tried the method proposed by Levinsohn and Petrin (2003) for the matched data. The results indicate that output value decreases by 0.0013% (p value = 0.11) with a 1% reduction in COD emissions. The estimated effect is much smaller than the baseline result.

The above checks suggest that our baseline results are robust whether we use a smaller sample with data for more recent years only, employ stricter criteria for linking the original datasets, use a two-year difference data sample, limit the sample to one-plant firms or long-surviving firms, use a stricter threshold to define regulation measure, or apply a fixed-effect model to the original matched data sample. All robust checks find that the water quality regulations had no statistically significant impacts on firms' emissions in the 3Rs3Ls basins. The checks also confirm that it would be costly for firms to reduce their COD emissions under the current production technologies.

8. Conclusions and implications

Using production and emission data for Chinese manufacturers, this paper evaluates the effects of the water quality regulations on surviving firms' COD emissions and productivity in the 3Rs3Ls basins in China. Empirical results suggest that China's regional strategy for controlling COD in the key regions had no statistically significant effects on firms' productivity. The underlying driver of this result is that the water quality regulations were ineffective in reducing surviving firms' COD emissions. Firms' output values would have been reduced should an effective regulation been imposed.

The empirical results must be interpreted with caveats. First, our data have limitations. The ASIF data do not include non-state-owned firms with sales less than 5 million RMB. The ESR data include only major emitters in each county. The exclusion of smaller firms may bias the productivity impact estimates downward because the water quality regulations often target smaller, inefficient firms. Furthermore, the present paper uses a dataset including only surviving firms when estimating the emission and production functions. The water quality regulations caused some small firms to shut down. Consequently, the effects of the water quality regulations on total emissions and output values may be larger than estimates from an analysis that covers only the surviving firms. However, due to data limitations, we could not delineate the effects of the regulations through its impacts on firm shutdowns.

Second, we use monetary value, not a physical measure, of a firm's output in the analysis of the effect of the water quality regulations on output. This might bias the results because the regulations may affect the total output, which may in turn affect the price of the output (Greenstone et al., 2012). For future research, the bias from these two sources should be examined when additional data become available.

[33] Changes in inputs of capital, labor, and intermediates are excluded from this particular regression.
[34] Table 4 summarizes the matched data.

Third, we find that the water quality regulations had no statistically significant effect on surviving firms' COD emission levels. This does not necessarily imply that the regulations have no impact on total emissions in the 3Rs3Ls basins. As pointed out above, many small plants with high COD intensity were forced to shut down in the 3Rs3Ls basins. The regulations then lead to reductions in total COD emissions from the industries without a significant impact on productivity of the surviving firms, which tend to be larger, more productive, and less pollution intensive.

The results from this analysis are useful for understanding the opportunity cost of further COD emission reductions from the targeted industries. As a national strategy for limiting emissions, the central government of China has implemented a policy often referred to as the "total quantity control of pollutants." The 11th Five-Year Plan (2006–2010) emphasized the importance of environmental protection in China's national development strategy and set specific goals for reducing major air and water pollutants, including a 10% reduction in sulfur dioxide (SO2) and COD emissions from the 2005 level (Liu and Diamond, 2008). The 12th Five-Year Plan (2011–2015) set up a goal of reducing COD emissions by 8% from the 2010 level.[35] The most recent Five-Year Plan (2016–2020) proposed to reduce COD emissions by another 10% from the 2015 emission level. Compliance with the COD reduction target becomes a necessary condition for local officials to be considered for promotion. Our results suggest that the opportunity cost for achieving the 10% COD emission reduction target is small in the 3Rs3Ls basins even under the current production technologies. Given that the manufacturing sectors are a major source of COD emissions, our estimates provide useful information for policy debates about the benefits and costs of water quality regulations in China.

Appendix

Table A1
COD emissions by industry (year 2004).

Industry (CIC Code)	Emissions	Share (%)
Food Processing from Agricultural Products(13)	602.42	13.34
Food (14)	146.52	3.24
Beverages (15)	198.8	4.40
Tobacco (16)	5.66	0.13
Textiles (17)	302.57	6.70
Textile Apparel, Footwear, and Caps (18)	14.87	0.33
Leather, Fur, Feather and Related Products (19)	68.59	1.52
Processing of Timber (20)	35.23	0.78
Furniture(21)	1.46	0.03
Paper and Paper Products (22)	1488.26	32.95
Printing, Reproduction of Recording Media (23)	2.02	0.04
Articles for Culture, Education and Sport Activity (24)	1.07	0.02
Processing of Petroleum, Coking, Processing of Nuclear Fuel (25)	64.9	1.44
Raw Chemical Materials and Chemical Products (26)	504.87	11.18
Medicines (27)	143.67	3.18
Chemical Fibers (28)	90.29	2.00
Rubber (29)	7.38	0.16
Plastics (30)	6.74	0.15
Non-metallic Mineral Products (31)	54.6	1.21
Smelting and Pressing of Ferrous Metals (32)	160.56	3.55
Smelting and Pressing of Non-ferrous Metals (33)	29.68	0.66
Metal Products (34)	14.72	0.33
General Purpose Machinery (35)	18.51	0.41
Special Purpose Machinery (36)	13.4	0.3
Transport Equipment (37)	60.36	1.34
Electrical Machinery and Equipment (39)	7.07	0.16
Communication Equipment, Computers and Other Electronic Equipment (40)	12.72	0.28
Measuring Instruments and Machinery for Cultural Activity and Office Work (41)	13.27	0.29
Artwork and Other Manufacturing (42)	2.55	0.06
Recycling and Disposal of Waste (43)	0.32	0.01
Manufacturing sum	4073.09	90.18
Industry (mining, manufacturing, and public utilities) Total	4516.63	100.00

Note: Data source is *China Environment Yearbook*. Emissions are measured in thousand metric tonnes.

[35] From 2010 to 2015, COD emissions actually decreased by 12.9%. The actual emissions in 2010 decreased by 12.5% compared to the 2005 level.

Table A2
Share of emissions of chemical oxygen demand by industry in the basins (%, 2002).

Industry (CIC Code)	Huai River	Liao River	Hai River	Tai Lake	Dianchi	Chao Lake
Food Processing from Agricultural Products (13)		7.0				
Food (14)	4.2		6.3			
Industries 13 and 14 combined				2.3		
Beverages (15)	5.9	10.5				
Tobacco (16)					5.4	
Textiles (17)	4.6			30.2	2.3	
Paper and Paper Products (22)	47.5	32.0	39.9	11.5		
Raw Chemical Materials & Chemical Products (26)	16.2	7.6	9.9	22.5		19.1
Medicines (27)		10.4	15.4		1.9	
Smelting and Pressing of Ferrous Metals(32)		7.7	4.3	6.6	83.4	39.2
Smelting and Pressing of Non-ferrous Metals (33)						10.0
Electrical Machinery and Equipment(39)						8.5
sum	78.4	75.2	75.8	73.1	93.0	76.8

Note: Data source for this table is *Annual Statistic Report on Environment in China 2002*, which was published by State Environmental Protection Administration.

Table A3
The ASIF data: Eight COD-emitting industries.

Year	# firms	Mean level of		
		Gross output	Capital stock	Employment
1998	47,490	48.31	30.83	360
1999	49,639	53.67	32.97	349
2000	48,963	63.29	34.19	335
2001	51,195	67.35	34.20	311
2002	53,982	75.70	34.05	294
2003	59,697	85.22	34.89	277
2004	82,525	78.91	29.57	219
2005	82,819	96.13	34.84	228
2006	91,491	106.11	36.92	216
2007	100,618	119.15	38.69	206

Note: Gross output are measured in million 2007 RMB.

Table A4
COD intensity regression.

	Dep. var.: COD intensity
Water quality regulation	−0.7534**
	(0.3436)
Aged 10 years or older	0.2235
	(0.2228)
With multiple plants	−0.4013
	(0.3693)
Export/sales value	−0.1849
	(0.4426)
Log(1 + county industry-level employment)	−0.4232***
	(0.0433)
Ownership dummies:	
COE	−2.0649***
	(0.4348)
DJV	−1.5547***
	(0.3004)
Private	−1.9603***
	(0.3888)
FIE	−2.0130***
	(0.4237)
HMTIE	−2.0950***
	(0.4363)
Regional dummies:	
Eastern	−2.2715***
	(0.3205)
Central	−0.4143
	(0.3576)
Size dummy:	
Large	−1.5362***
	(0.2182)
N	42,166

Notes: 1. COD intensity is measured in tonnes per million (2007) RMB. The regression controls for a constant and 4-digit industry by year fixed effects. 2. For the ownership dummies, the base category is state-owned enterprise. 3. See Fig. 1 for the classification of three regions. 4. A firm is treated as a large one in a year if its output value is greater than the mean in the ASIF industry-year group.

Table A5
The effects of environmental regulation on COD emissions: more results.

	Dep. var.: Δlog (COD emission)					
	(1)	(1)'	(2)	(2)'	(3)	(3)'
Water quality regulation	0.0137	−0.0115	−0.0212	−0.0580*	0.0003	−0.0682
	(0.0282)	(0.0254)	(0.0374)	(0.0327)	(0.0495)	(0.0426)
Water quality regulation*log(COD intensity)	−0.0169	0.0177	−0.0217	−0.0545	−0.0173	−0.0385
	(0.0127)	(0.0338)	(0.0162)	(0.0462)	(0.0205)	(0.0597)
Firm characteristics	Y	Y	Y	Y	Y	Y
4-digit industry by year FE	Y	Y	Y	Y	Y	Y
County FE			Y	Y		
County by year FE					Y	Y
N	15,193	15,193	14,808	14,808	11,213	11,213

Note: Columns (1)', (2)', and (3)' report IV estimates for the three specifications (1)–(3), respectively. See Table 7 for results from regressions controlling for firm fixed effects.

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